Multiple IV Bolus Practice 1

Module 6 - Multiple Dosings · Daily practice sheet · No date

Name _____ Date _____

1a One gram of a drug was administered by IV bolus injection every 8 hours to a 65-kg patient. The drug has an apparent V_D of 26.2 L/100 kg. The total body clearance is 52.95 mL/min per 70 kg and renal clearance is 34.72 mL/min per 70 kg. Calculate the elimination half-life.

1b One gram of a drug was administered by IV bolus injection every 8 hours to a 65-kg patient. The drug has an apparent V_D of 26.2 L/100 kg. Calculate the maximum plasma drug concentration of the first dose.

1c One gram of a drug was given by IV bolus every 8 hours. The first dose peaks at 58.72 mg/L and k is 0.1732 hr^{-1}. Calculate the minimum plasma drug concentration of the first dose.

1d One gram of a drug is given by IV bolus every 8 hours. The first dose peaks at 58.72 mg/L and k is 0.1732 hr^{-1}. Calculate the maximum steady-state plasma drug concentration.

1e A drug given by IV bolus every 8 hours reaches 78.31 mg/L at its steady-state peak, and k is 0.1732 hr^{-1}. Calculate the minimum steady-state plasma drug concentration.

1f One gram of a drug is given by IV bolus every 8 hours to a patient whose total body clearance is 2.95 L/hr. Calculate the average steady-state plasma drug concentration.

1g A drug given by repeated IV bolus has an average steady-state concentration of 42.37 mg/L and an apparent V_D of 17.03 L. Calculate the average amount of drug in the body during steady state.

1h A drug given by IV bolus every 8 hours has a steady-state peak of 78.31 mg/L and k of 0.1732 hr^{-1}. Calculate the plasma level 12 hours after the last dose, assuming steady-state levels are achieved.